

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented moderately complex integrated circuit layouts using **EDA tools**, contributing to the development of advanced converter technologies.
- Performed **timing extraction and analysis** with increasing independence on integrated circuit designs, ensuring adherence to performance specifications.
- Applied foundational knowledge of **IC layout design principles** and **CMOS technology** to resolve moderately complex layout challenges.
- Coordinated with cross-functional teams, including **process engineers** and design leads, to integrate layout designs into semiconductor manufacturing workflows.
- Contributed to **short-term focused projects** with clear deliverables, such as layout verification and optimization tasks, under moderate guidance.
- Gained proficiency in using **EDA tools** for layout design, simulation, and verification, improving efficiency in design cycles.
- Analyzed technical data and design specifications to identify potential issues and propose layout adjustments, demonstrating an understanding of how individual work impacts business objectives.

SKILLS

Layout Fundamentals – IC layout design principles, CMOS technology, semiconductor manufacturing processes, timing analysis, EDA tools

Technical Skills – Integrated circuit layout, layout verification, timing extraction

Software Tools – EDA tools

Methodologies – Agile methodologies, Scrum ceremonies

Business Skills – Business objectives alignment, cross-functional team coordination

PROJECTS

Integrated Circuit Layout Design for Advanced Converters

- Designed and implemented integrated circuit layouts of moderate complexity for advanced converter applications, focusing on critical performance parameters.
- Conducted detailed timing analysis and extraction using **EDA tools** to ensure designs met stringent performance requirements for high-speed applications.
- Collaborated with **semiconductor manufacturing process** engineers to validate layout designs against fabrication constraints and optimize for yield.

Layout Optimization and Verification Framework

- Developed a systematic approach for layout verification, ensuring adherence to design rules and electrical requirements for complex ICs.
- Applied **CMOS technology** principles to optimize layout density and reduce parasitic effects, improving overall circuit performance.
- Assisted in the analysis and resolution of layout-related issues encountered during the design process, contributing to project timelines.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)